

Project Ember — The AI HoneyToken Factory

A cybersecurity deception platform for African SMEs that uses Google Gemini to generate realistic decoy documents that trap ransomware and malicious insiders.

The Problem

African SMEs face the same threats as global enterprises — ransomware, insider data theft, and targeted attacks — but lack the budget for enterprise-grade deception tech (e.g., Illusive Networks, Attivo). When ransomware hits, it's often too late: files are encrypted, data is exfiltrated, and there are no logs to trace the source.

How Project Ember Solves It

Project Ember flips the asymmetry: instead of trying to block every attack, you **plant decoys** that look like high-value targets (payroll spreadsheets, HR records, IT credentials) across your file servers. When an attacker touches one — whether it's ransomware encrypting files, or an insider browsing sensitive docs — the dashboard alerts you in real time.

The Flow

1. DEPLOY —> Gemini generates 3 realistic decoy files (.txt, .md, .html) with hidden tracking pixels and fake corporate data (payrolls, employee records, IT audit reports)
2. PLACE —> Copy the files to your file server, shared drive, or sensitive folder where attackers would look
3. WAIT —> The dashboard monitors:
 - Tracking pixel hits (file opened / viewed via browser)
 - Filesystem changes (modified / deleted / renamed)
 - All via the watchdog daemon
4. ALERT —> When a trap is triggered:
 - Full-screen red banner with file name + timestamp
 - Sky News-style audible alarm
 - Graph node turns red with pulsing animation
 - 📊 Event counter updates in real time

Current Capabilities

Feature	Status
AI Decoy Generation — Gemini 2.0 Flash generates convincing corporate documents with African context (KES/NGN currencies, KRA/FIRS compliance, local names, East African logistics routes)	✓
4 Department Templates — Finance (payroll), HR (employee records), IT (network audit), Operations (supply chain)	✓
3 Output Formats — <code>.txt</code> , <code>.md</code> , <code>.html</code> per deployment	✓
Hidden Tracking Pixels — <code></code> beacon in HTML/MD files; logs hits to the dashboard	✓
Filesystem Watchdog — detects MODIFIED, DELETED, MOVED, CREATED events in real time	✓
Network Topology Graph — tree-layout visualization (server → departments → decoy files) using networkx + matplotlib	✓
Live Threat Monitor — event log with timestamps, types, and file names	✓
Explosive Alert — red banner at page top with event details + <code>st.error()</code>	✓
Flashing Red Nodes — graph nodes pulse red when a file is accessed	✓
Audible Alarm — Sky News–style two-tone alert (bass thump + rising sweep) via Web Audio API	✓
No-Refresh Polling — background poller checks every 4s without page flicker	✓
Bootstrap Icons — professional UI, clean aesthetics	✓
Mock Mode — works without API key for UI testing (shows placeholder text)	✓

Tech Stack

Layer	Technology
AI Engine	Google Gemini 2.0 Flash via <code>google-genai</code> SDK
Dashboard	Streamlit 1.60+
Graph Visualization	networkx + matplotlib
Fake Data	Faker (custom African name pool, IBANs, ID numbers)
File Monitoring	watchdog (filesystem event listener)
Tracking Server	Python <code>http.server</code> (lightweight, returns 1×1 GIF pixel)
Alert Sound	Web Audio API (generated in-browser, no audio files)
Icons	Bootstrap Icons CDN
Background Tasks	Python threading

How to Use

1. Install

```
cd Project_EMBER
pip install -r requirements.txt
export GEMINI_API_KEY="your_google_ai_key_here"
```

2. Launch

```
python3 run.py
```

This starts three services:

- **Tracking server** on **:8765** — handles beacon pixel requests
- **Watchdog daemon** — monitors **./honeytokens/** for file changes
- **Streamlit dashboard** on **:8501** — the UI

3. Deploy Decoys

1. Open **http://localhost:8501** in a browser
2. Enter a **Company Name** (e.g., "Acme Kenya Ltd")
3. Pick a **Department** (Finance / HR / IT / Operations)
4. Click **Deploy HoneyTokens**

Three files appear in **./honeytokens/**:

- **Company_Dept_20260724.html** — open this in a browser to test tracking
- **Company_Dept_20260724.md**
- **Company_Dept_20260724.txt**

4. Place the Traps

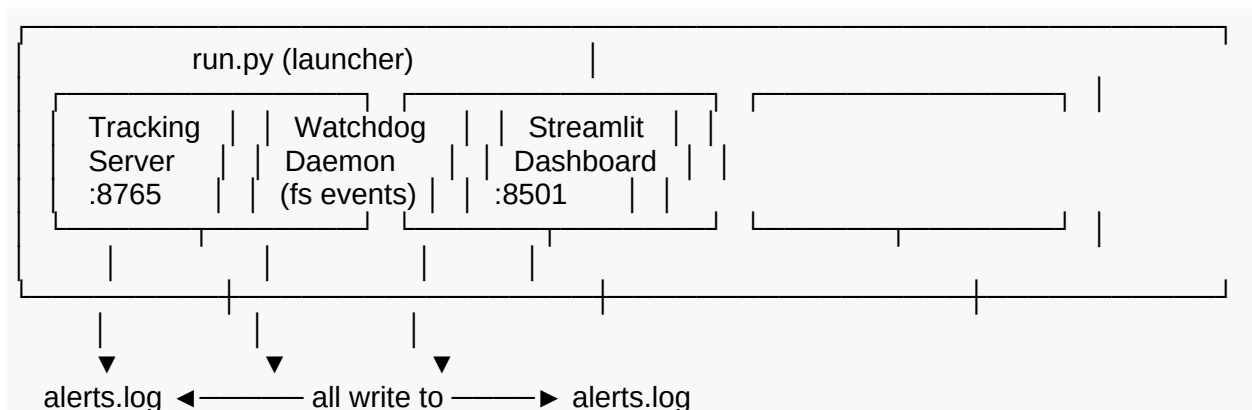
Copy the generated files to your target environment — a file server, a shared drive, an HR document folder, or anywhere an attacker might browse.

5. Monitor

Keep the dashboard open. When someone (or something) touches a decoy:

Action	What the dashboard shows
Opens .html in a browser	Tracking pixel fires → red banner + sound + graph node turns red
Modifies a file (ransomware)	Watchdog catches → same alert + event log entry
Deletes/moves a file	Watchdog catches → alert + log

Architecture



↓
Generator (Gemini + Faker) —▶ ./honeytokens/

File Structure

```
Project_EMBER/
├── app.py          # Streamlit dashboard (graph, alerts, metrics)
├── generator.py    # Gemini + Faker decoy generation
├── watchdog_daemon.py # Filesystem event monitor
├── run.py          # Launcher (tracking + watchdog + streamlit)
├── requirements.txt # Python dependencies
├── honeytokens/    # Generated decoy files
│   ├── manifest.json # Tracking manifest
│   ├── *.html        # Decoy with pixel beacon
│   ├── *.md
│   └── *.txt
```

Deploying for Real

For production use:

- **Reverse proxy** (nginx/Caddy) to serve both Streamlit and tracking on port 80/443
- **Persistent alerts.log** with rotation
- **Watchdog** on the actual file server path (not local `./honeytokens`)
- **SSL** for all endpoints
- **Email/Slack/Webhook** integration for off-hours alerts